
Nathan Seidle

SparkFun Electronics

Combos in 45

minutes or less!*

*You know what these mean...

2002



New Products

Top Sellers

SparkFun Originals

Spectacle

Sale

Gift Certificates

Arduino

Audio

Books

Breakout Boards

Cables

Components

Development Tools

Dings and Dents

Educators

GPS

Intel® Edison

IoT

Kits

LCDs

Prototyping

Raspberry Pi

Robotics

Sensors

SparkX

Swag

Tools

Way more versatility than
just radio, the bladeRF x 40.

EXPLORE THE REAL RADIO STAR



New Products

See All

PAGE 5 OF 6



SparkFun micro:climate
kit

★ KIT-14217
~~\$114.95~~ \$54.95



SparkFun micro:arcade
kit

★ KIT-14218
~~\$49.95~~ \$46.78



SparkFun moto:bit

★ DEV-14213
~~\$14.95~~ \$11.18



SparkFun weather:bit

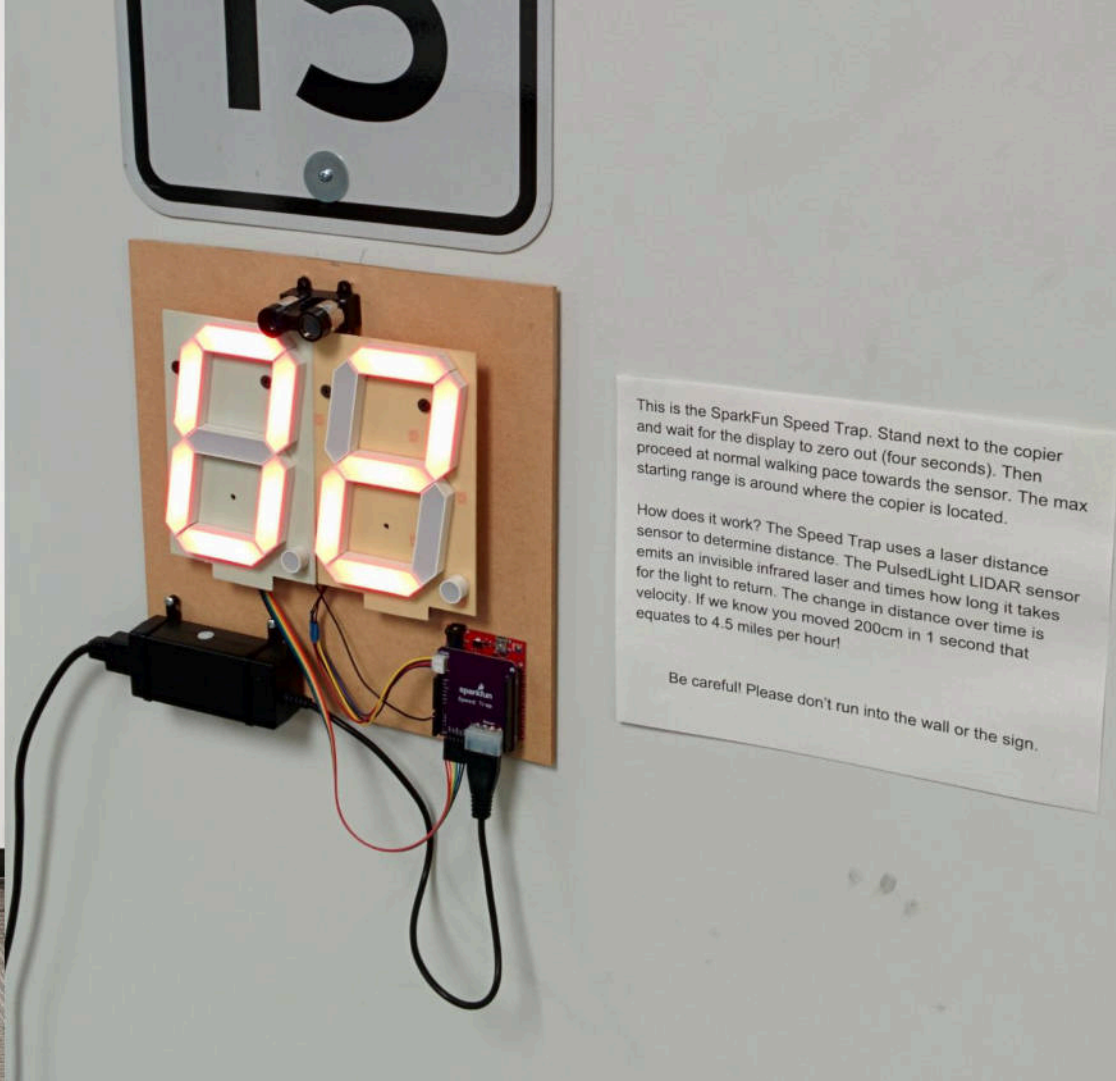
★ DEV-14214
~~\$14.95~~ \$10.95

LATEST BLOG POST

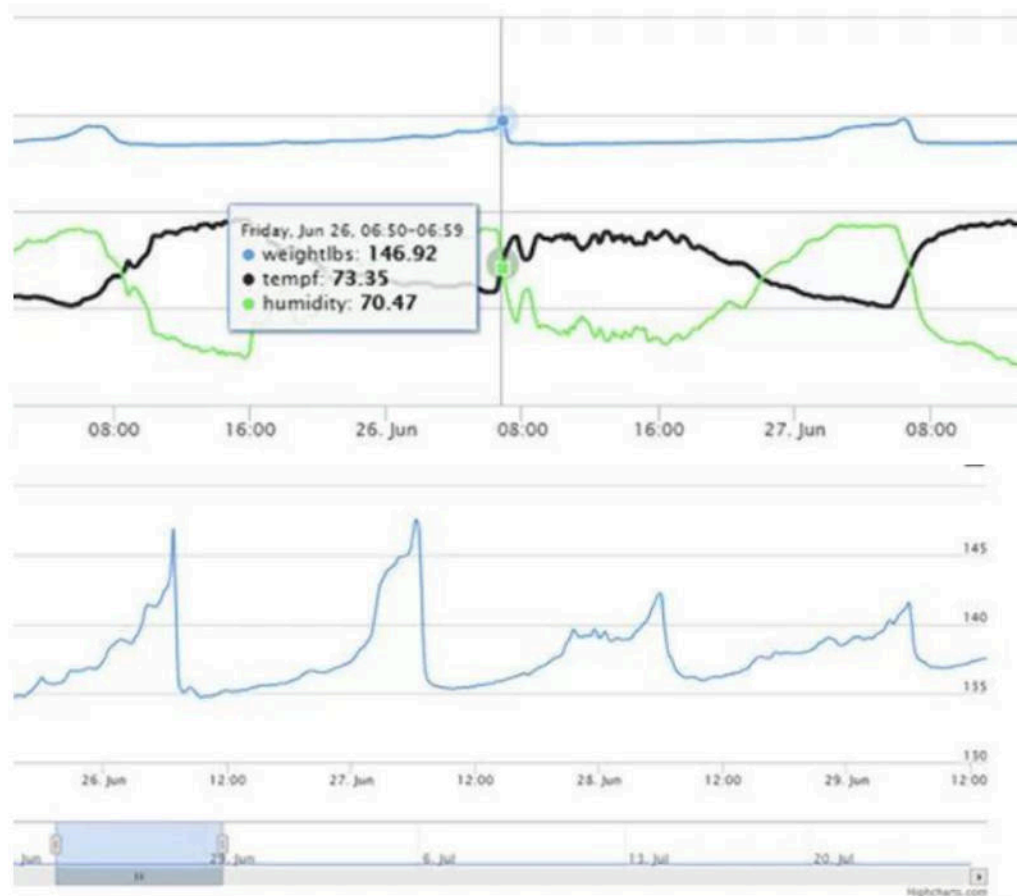
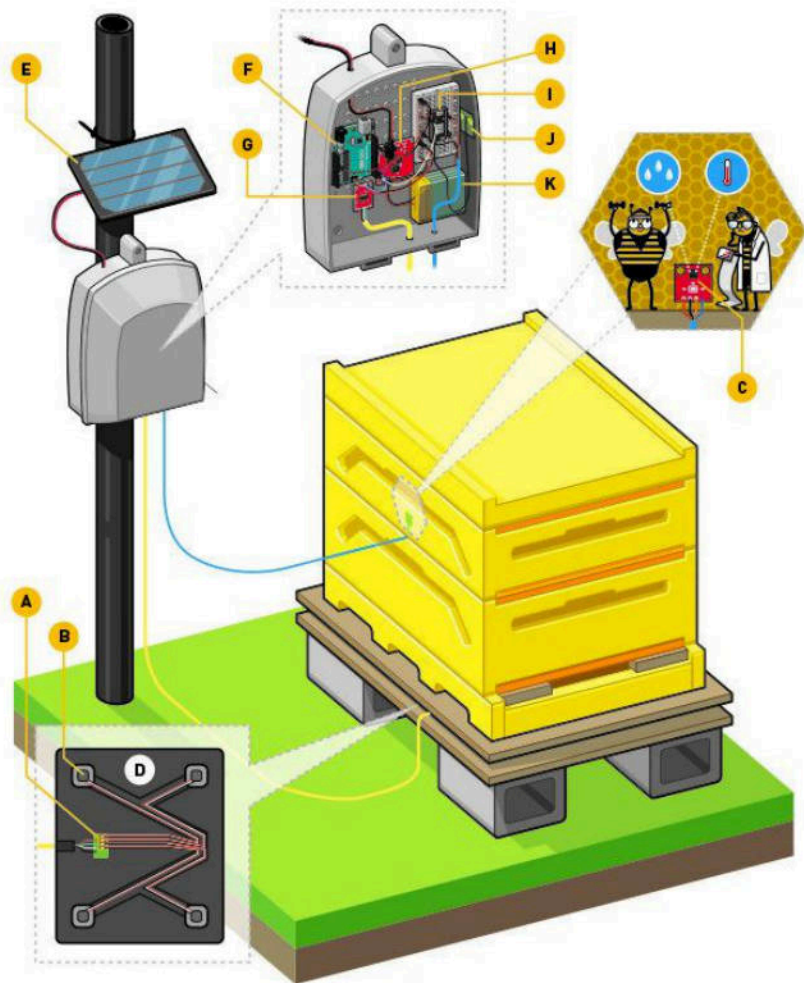
Desk of an Engineer: the Mary Edition



FREE
SHIPPING



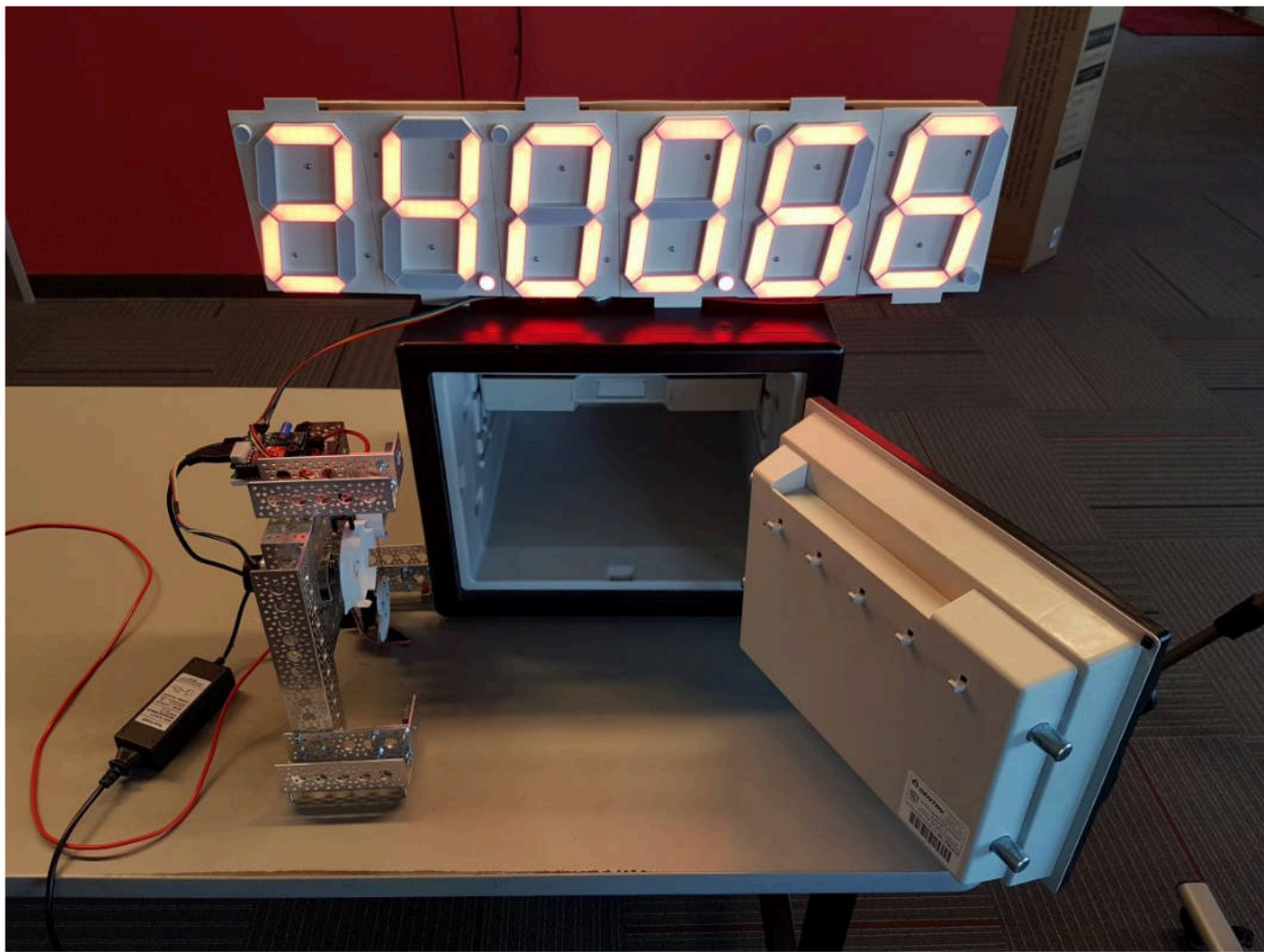






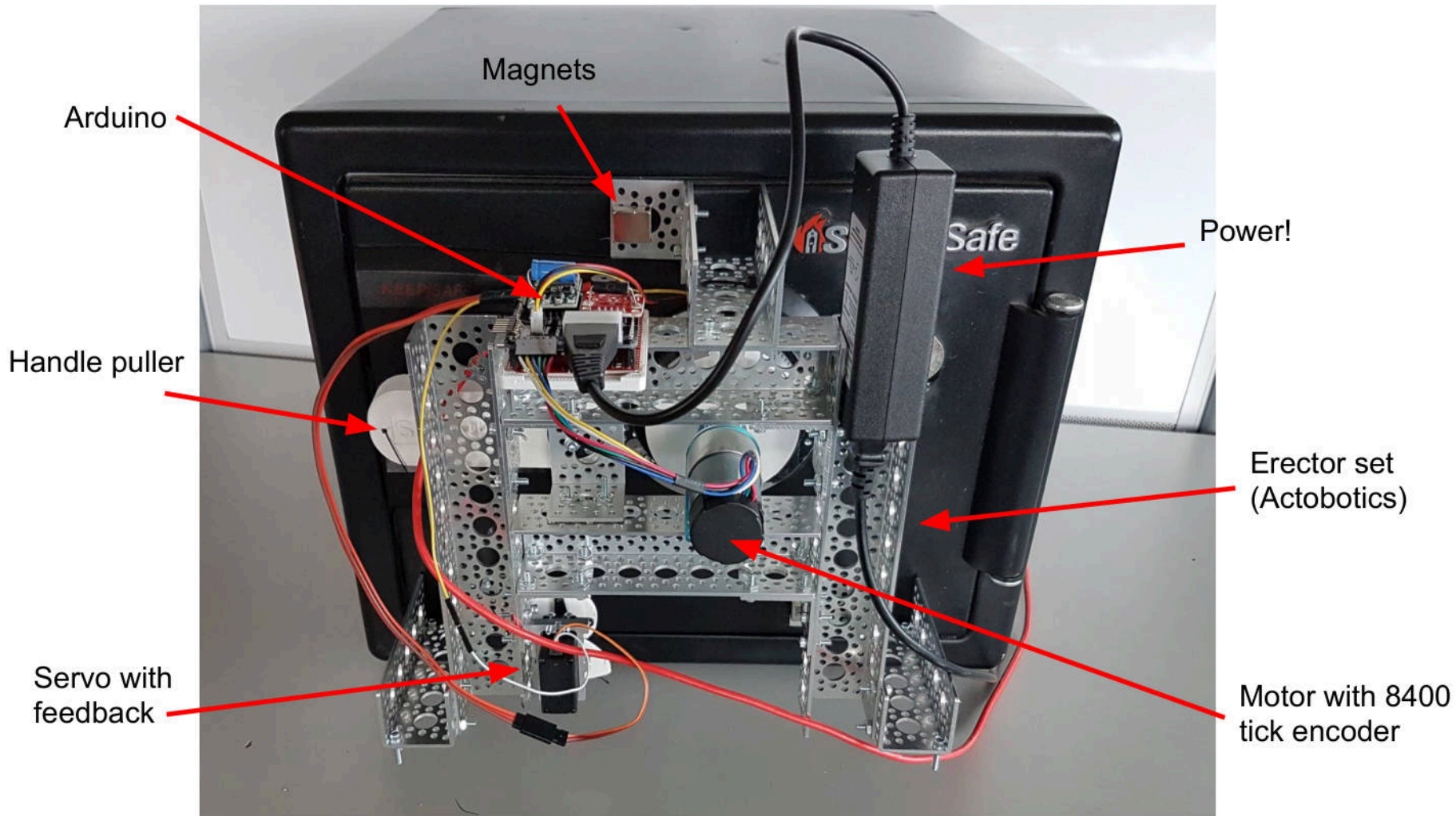


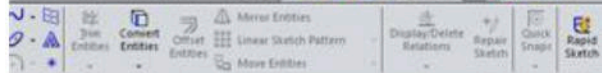




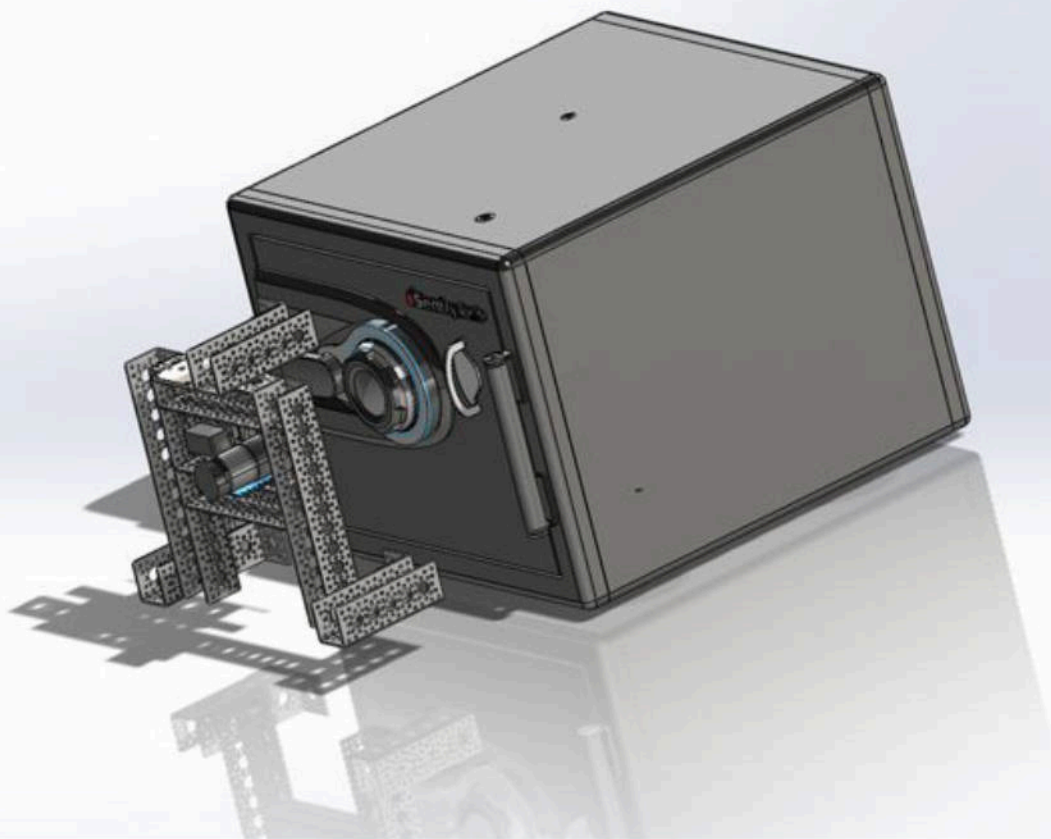
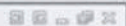
Animation:

How combination safe works





multigert Office Products

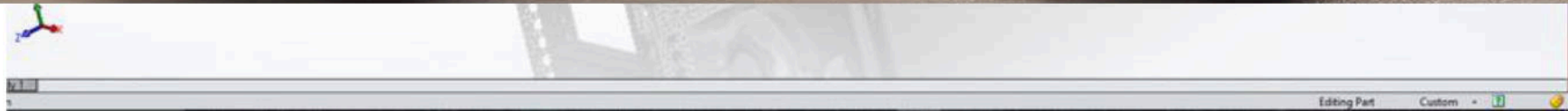
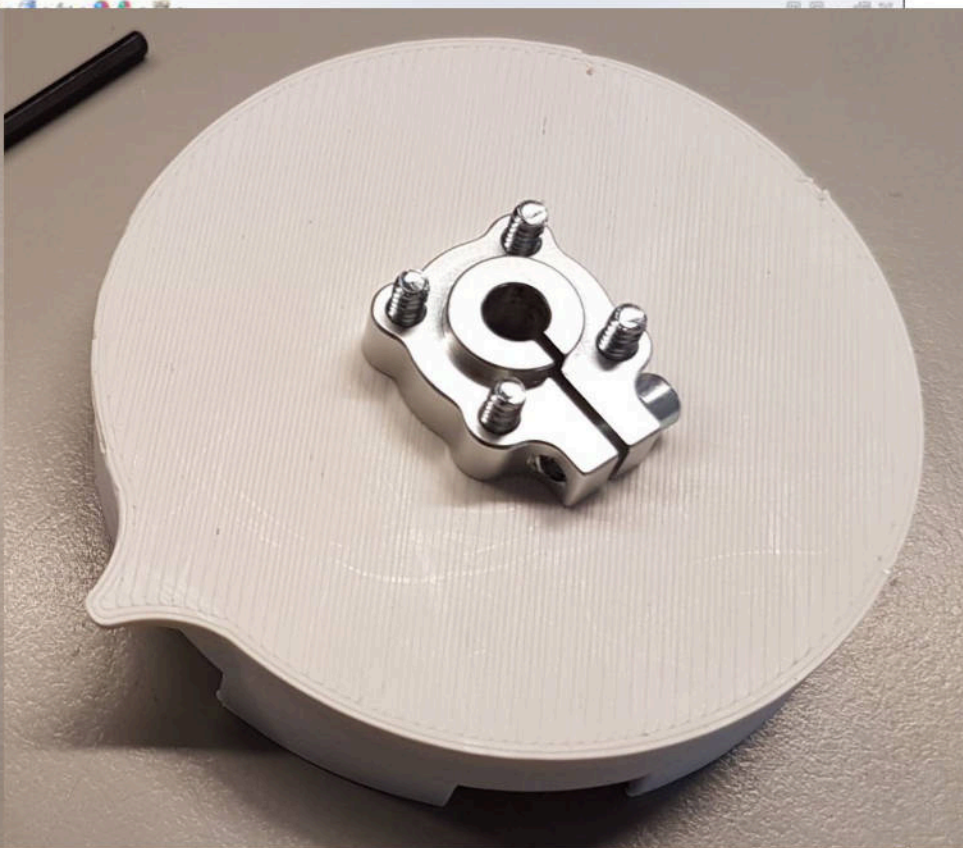
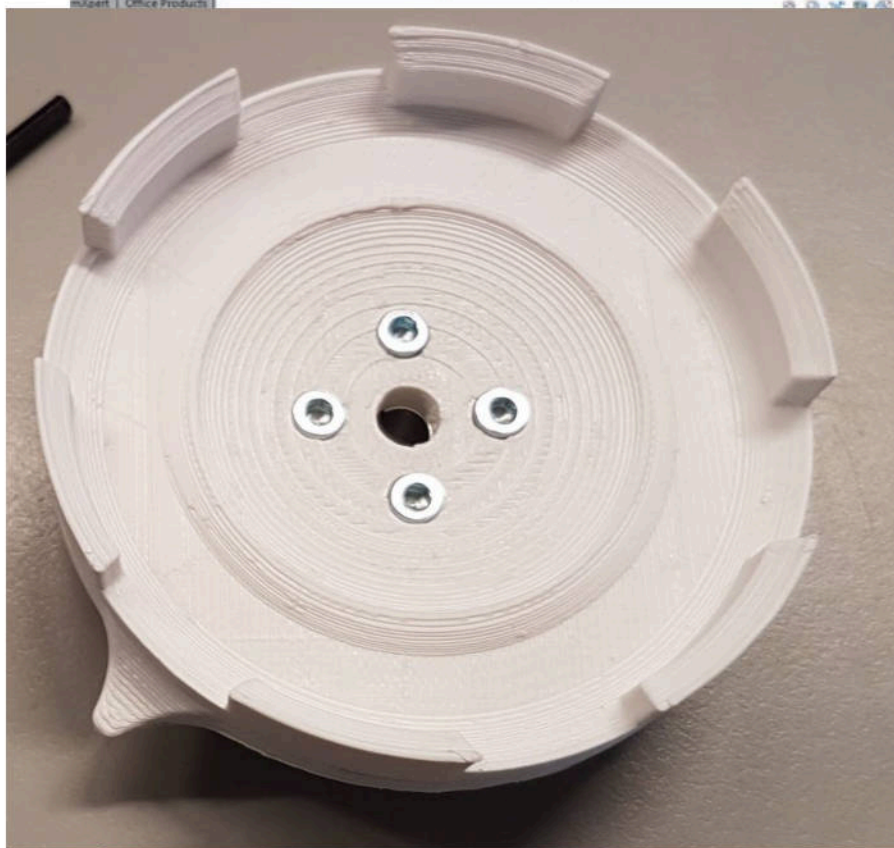
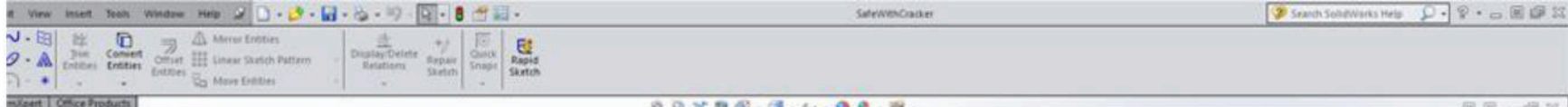


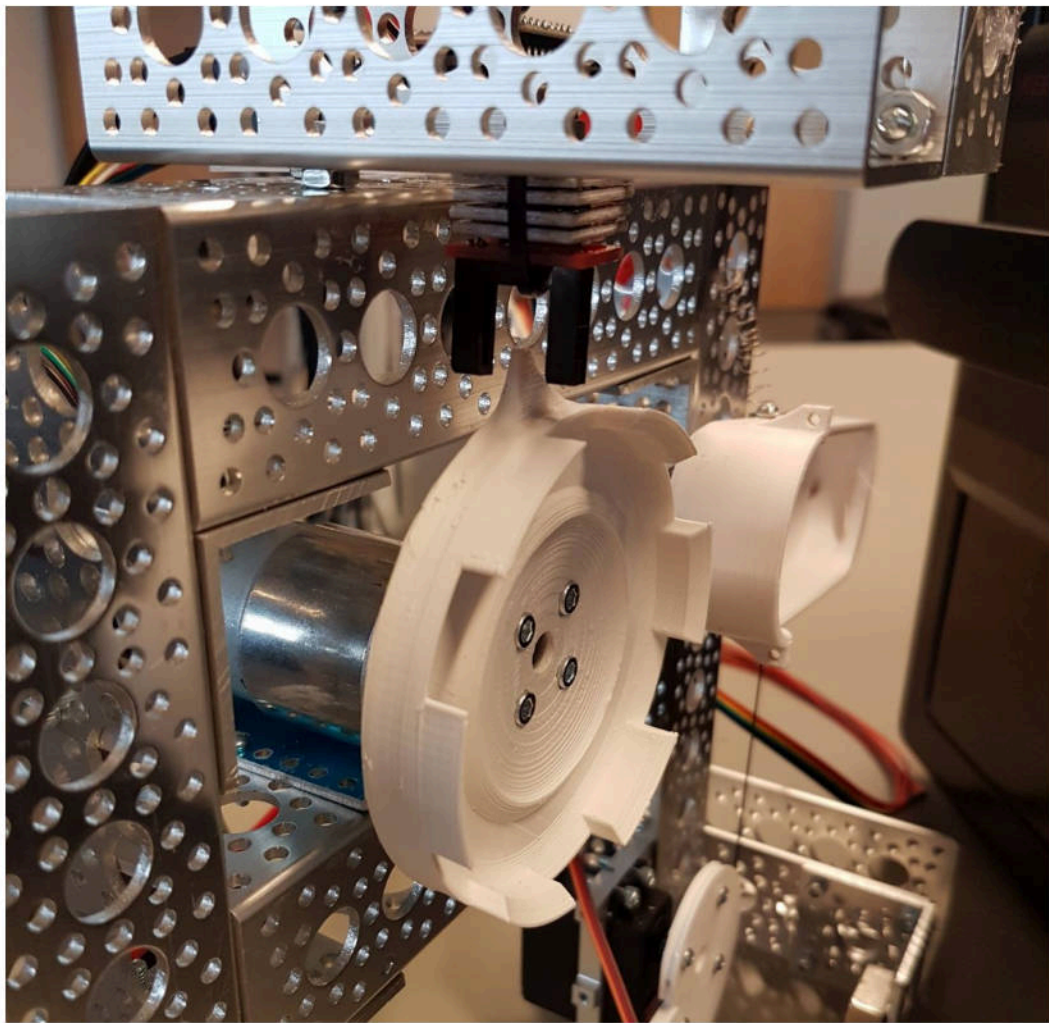
9.1.1

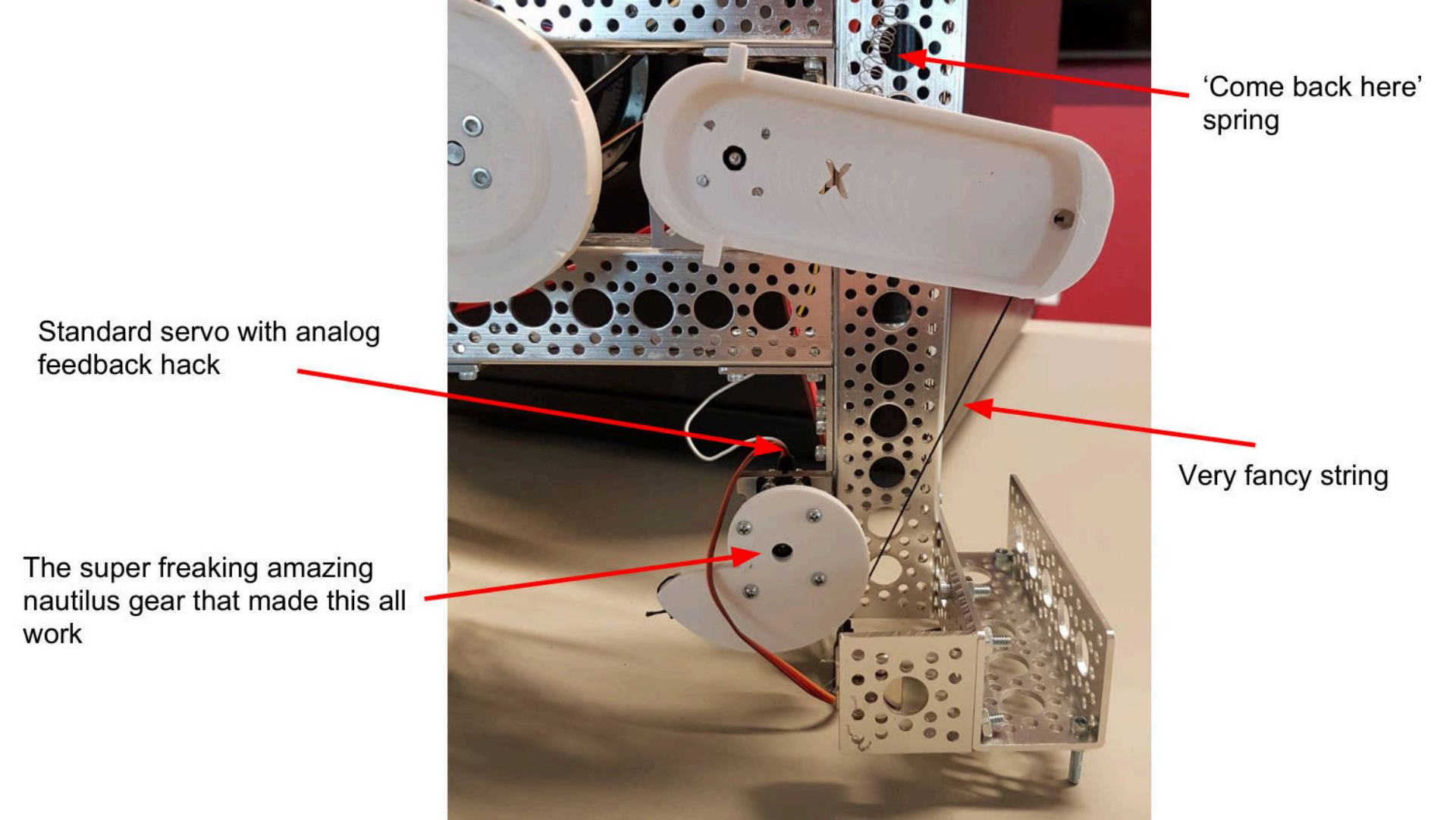
Editing Part

Custom







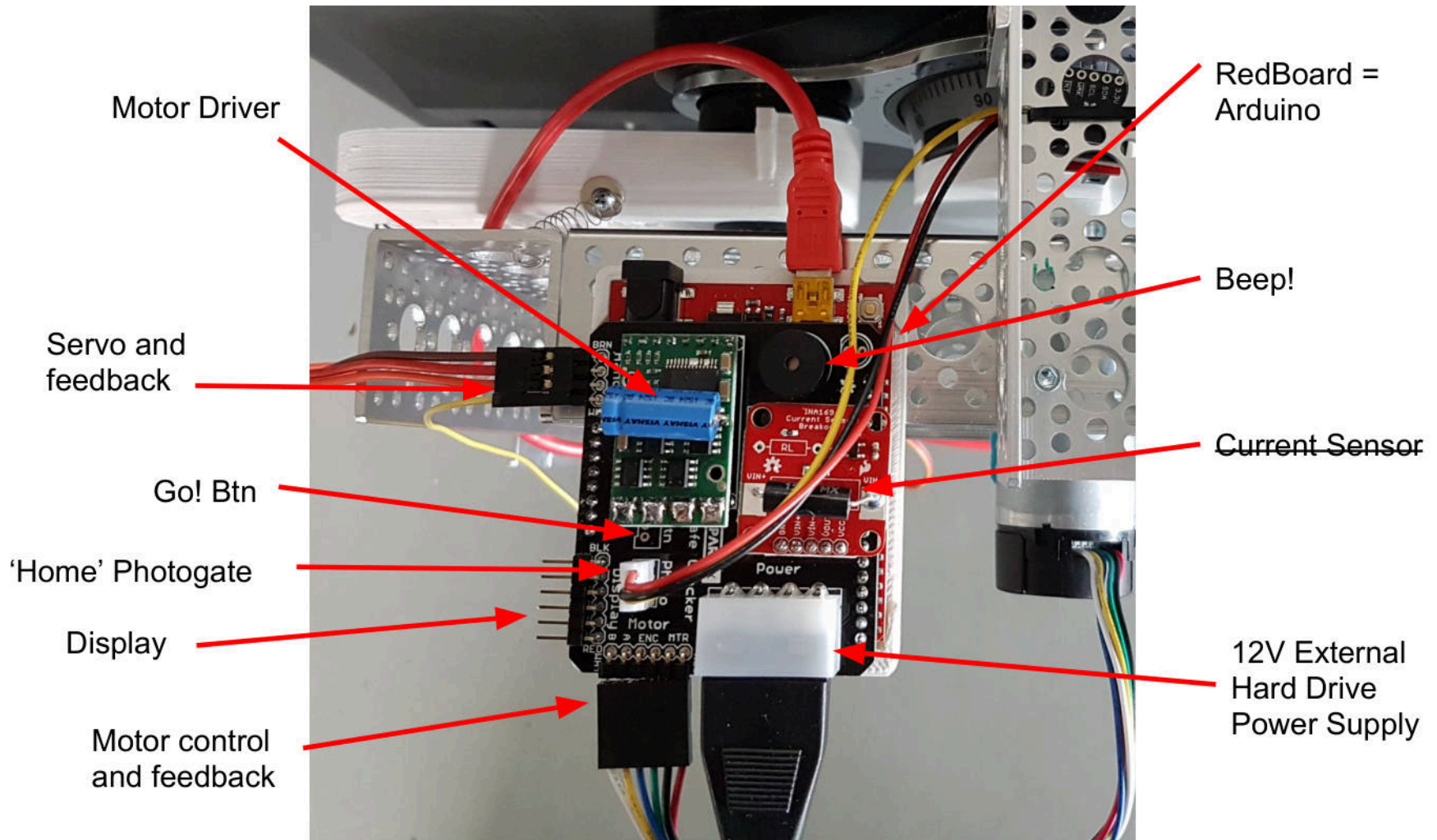


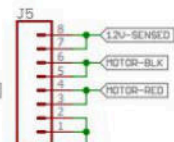
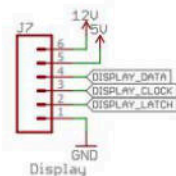
'Come back here'
spring

Standard servo with analog
feedback hack

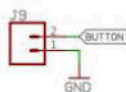
Very fancy string

The super freaking amazing
nautilus gear that made this all
work

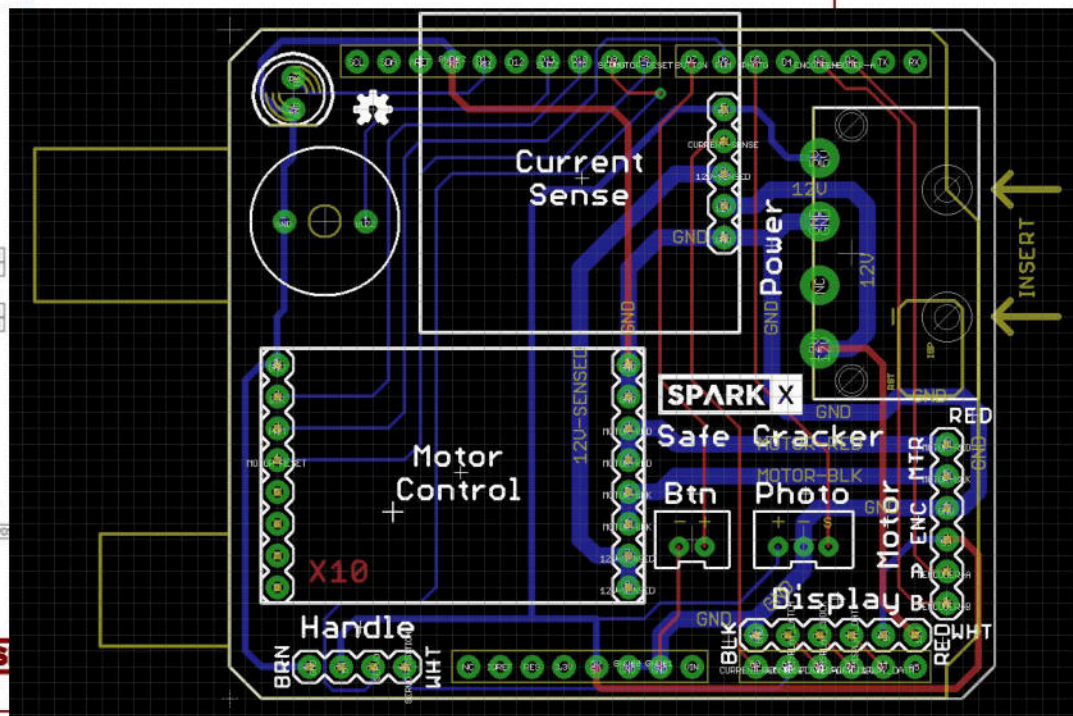
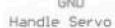




Motor Controller Output



Sheet: 1/1



Problem Domain:

100^3 combinations

10 seconds per test

115 days (worst case)

Combinations:
 100^3 combinations

Exploits

Combinations:

~~**100^3 combinations**~~

33^3 combinations = 4.15 days

Exploits

Combinations:

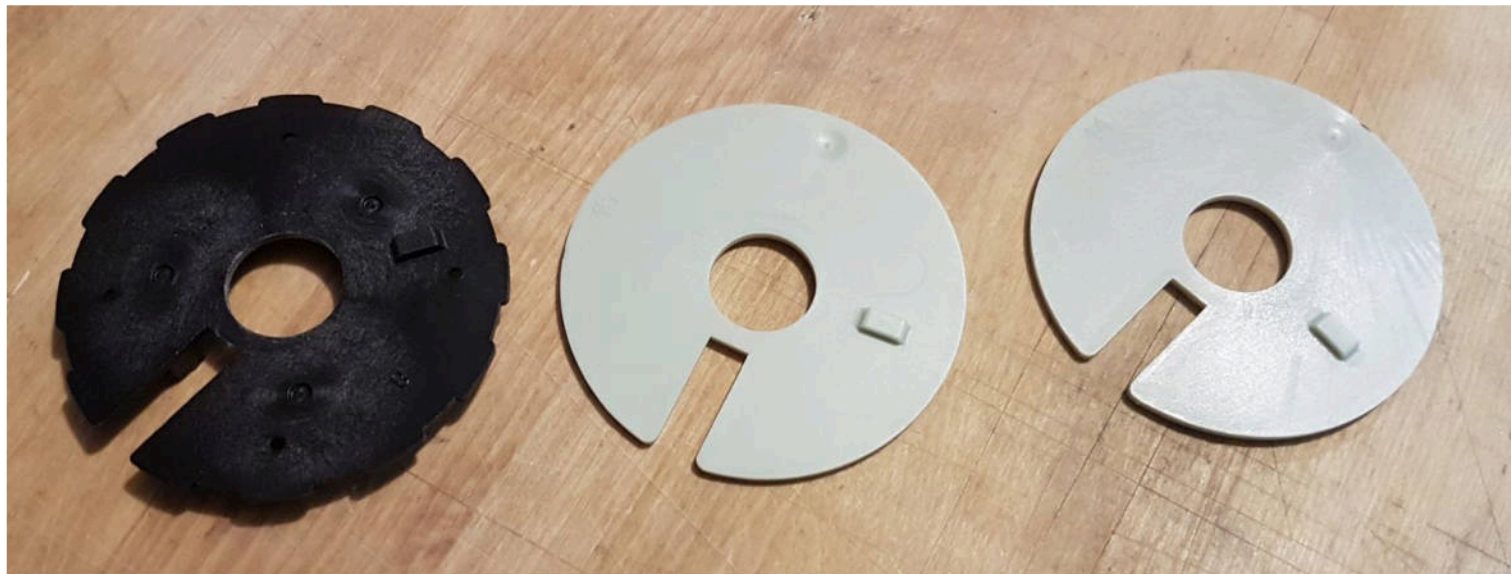
~~**100^3 combinations**~~

~~**33^3 combinations = 4.15 days**~~

Disc C has 12 indents

$33^2 * 12 = 1.5$ days

Exploits



Exploits



Disc C:

Outer diameter: 2.815" (55.5mm)

Width of solution slot: 0.239"

Width of 11 indents: 0.249" +/- 0.002"

17.69" (Circumference) / 8400 ticks

0.0021" / tick

~5 ticks smaller

Exploits

Combinations:

~~**100^3 combinations**~~

~~**33^3 combinations = 4.15 days**~~

~~**Disc C has 12 indents**~~

~~**$33^2 * 12 = 1.5$ days**~~

Disc C has a skinny indent

$33^2 * 1 = 3$ hours

Exploits

Test Time:

Resetting Dials = 10s / test

Exploits

Test Time:

~~**Resetting Dials = 10s / test**~~

'Set testing' = 4s / test

1.2 hours

Exploits

Test Time:

~~Resetting Dials = 10s / test~~

~~'Set testing' = 4s / test~~

~~1.2 hours~~

45 minutes!

~~Exploits~~ **Luck**

Animation:

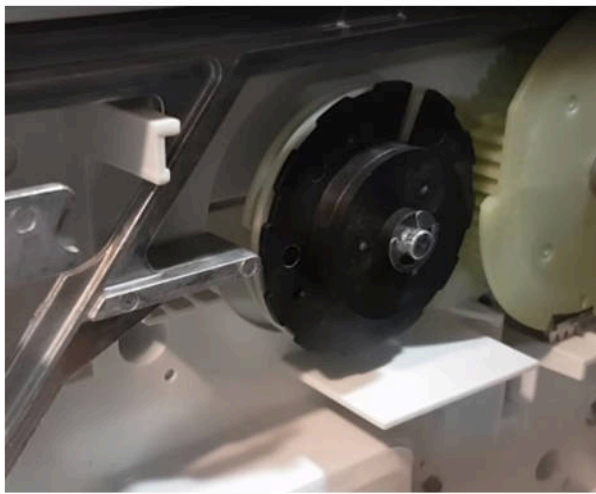
How Set Testing Works



How do I protect myself!?



One of these is not like the others...



The S&G 6741 ... can be dialed up to 1.25 digits above or below the actual set number and still open, essentially giving you a 2.5 digit window to hit. The S&G 6730, however, has only a $\pm .5$ dialing tolerance, essentially giving a 1 digit window to hit. While many locksmiths might prefer the S&G 6730, it can be notoriously difficult to open and very unforgiving to human error. In addition, slight alterations to the lock (for example, if the dial or the dial ring was bumped during shipping) can shift the combination, rendering the lock unusable.

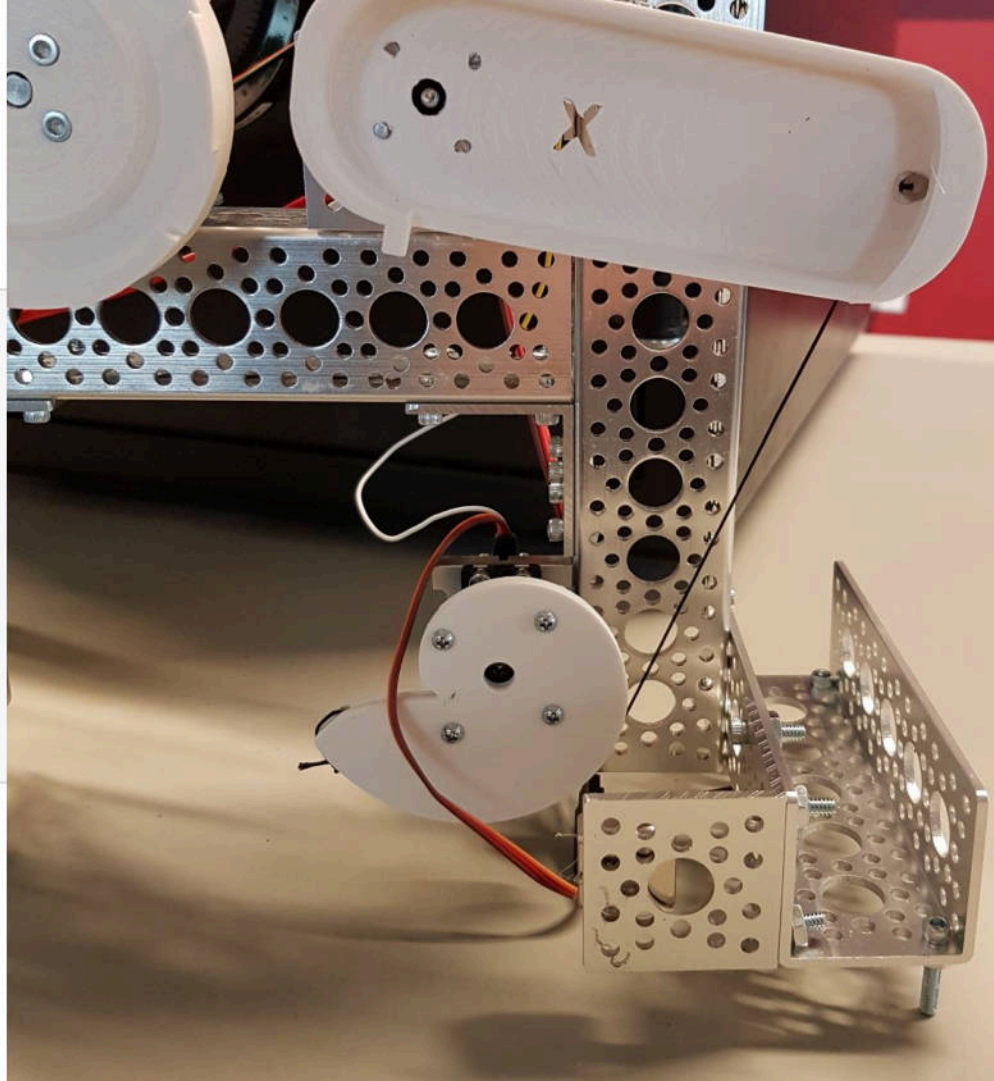




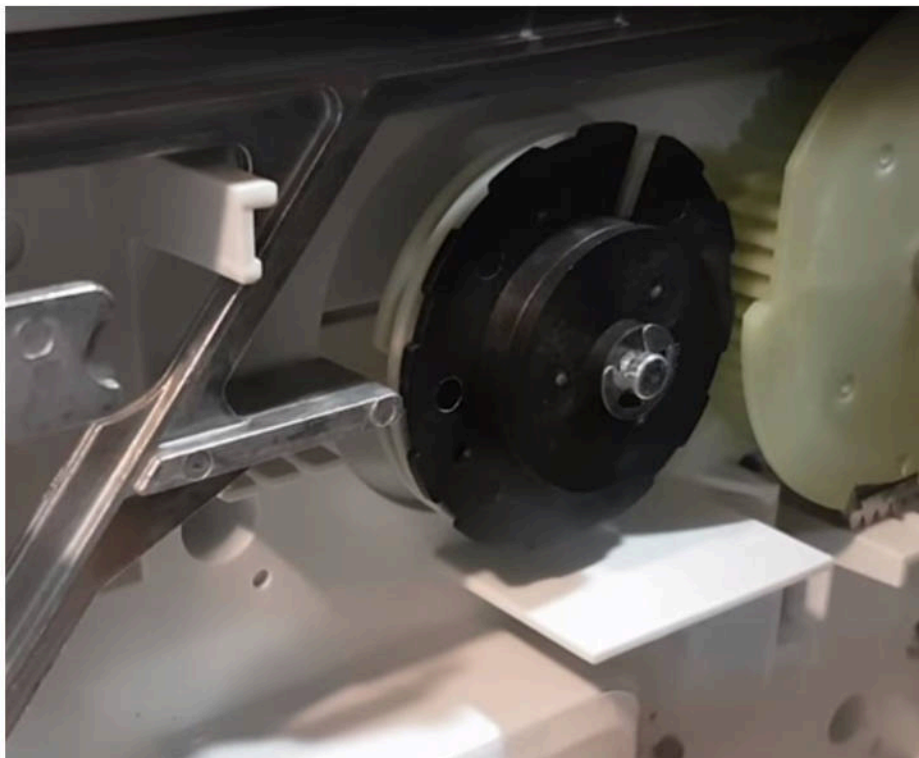
to 1.25 digits
er and still
it window to
only a $\pm .5$
g a 1 digit
might prefer
difficult to
n error. In
for example,
ed during
rendering the

Future Research

```
Measuring complete  
Smallest to largest width [Width]/[Depth]  
Indent 8: [1911] / [1130]  
Indent 1: [1925] / [1122]  
Indent 3: [1953] / [1091]  
Indent 0: [1955] / [1099]  
Indent 11: [1966] / [1105]  
Indent 2: [1992] / [1100]  
Indent 9: [1994] / [1126]  
Indent 7: [2011] / [1098]  
Indent 10: [2036] / [1096]  
Indent 4: [2077] / [1109]  
Indent 5: [2083] / [1100]  
Indent 6: [2114] / [1096]
```



Future Research



Future Research



Is it open yet?



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